Information Systems

Chapter 10

E-Commerce: Digital Markets, Digital Goods

1. What is E-Commerce?

E-commerce means buying and selling goods or services using digital networks, mainly the Internet.

> *E-commerce is digitally enabled commercial transactions between organizations and individuals.*

- Buying a book from Amazon
- Ordering food from Foodpanda
- Watching Netflix through subscription
- Buying software online
- Booking a ride through Uber
- Paying bills through a mobile app

2. E-Commerce vs Traditional Commerce

Dimension	Traditional Commerce	E-Commerce
Channel	Physical shops, markets	Website, mobile app, social media
Payment	Cash at counter	Online payment, mobile wallet
Reach	Local or regional	Global
Hours	Fixed business hours	24/7 availability
Interaction	Face-to-face salesperson	Digital chat, reviews, AI bots

3. The New E-Commerce: Social, Mobile, Local

Modern e-commerce has evolved beyond websites. Businesses now sell through social media, mobile apps, and location-based services.

3.1 Social E-Commerce

Uses social platforms to market and sell products. Key ideas: **conversation, engagement, empathy,** and **interaction** with customers.

- Facebook shops and Instagram stores
- TikTok product selling
- Influencer marketing
- Reviews, comments, and recommendations

EXAMPLE

A clothing brand posts a new design on Instagram. Customers comment, ask prices, and place orders through DM.

3.2 Mobile E-Commerce

Buying and selling through smartphones and tablets. Especially useful for location-based services, mobile payments, banking, and digital content downloads.

- Daraz mobile app
- Easypaisa / JazzCash payments
- Foodpanda app
- Mobile banking

3.3 Local E-Commerce

Uses customer location to provide nearby offers and services. Smartphones make geoadvertising, geosocial, and geoinformation services possible.

- Restaurants near me
- Careem showing nearby drivers
- Food delivery apps showing restaurants in your area
- A shop sending a discount to people nearby

4. Online Marketing

Online marketing means promoting products or services using digital channels. The Internet gives marketers new ways to identify and communicate with millions of potential customers at lower cost than traditional media.

Method	Example
Website	Company website showing products
Display Ads	Banner ads on websites
Email Marketing	Sending offers to customers
Video Ads	YouTube ads
Repeat Ads	Retargeting customers who visited before
Social Media Marketing	Facebook, Instagram, LinkedIn campaigns

5. Eight Unique Features of E-Commerce Technology

EXAM KEY TOPIC: These eight features explain why e-commerce is fundamentally different from traditional commerce.

5.1 Ubiquity

E-commerce is available everywhere and all the time. Traditional commerce requires going to a physical shop. E-commerce allows shopping from home, office, university, or any country. This creates a **marketspace** and reduces transaction costs because customers do not need to travel.

EXAMPLE

You can order a product from Daraz at 2 AM from your phone.

5.2 Global Reach

E-commerce can cross national and cultural boundaries. Traditional business is usually local or regional. E-commerce allows access to global customers more conveniently and cost effectively than traditional commerce.

EXAMPLE

A seller in Pakistan can sell digital services to a client in the USA through Fiverr or Upwork.

5.3 Universal Standards

The Internet uses common technical standards across the world. This reduces cost because businesses do not need different systems for every country. Advantages include lower market entry cost and lower search cost.

EXAMPLE

A website made in Pakistan can be opened in UAE, USA, or Saudi Arabia using the same Internet standards.

5.4 Richness

E-commerce can deliver rich information using text, images, audio, video, animation, reviews, and live chat. This makes online selling more powerful than simple printed ads.

EXAMPLE

A product page on Amazon shows images, videos, customer reviews, ratings, FAQs, and recommendations.

5.5 Interactivity

Two-way communication between buyer and seller. Unlike traditional broadcast marketing, e-commerce allows real-time dialogue.

EXAMPLE

A customer messages a brand on Instagram asking for size, price, and delivery details. The brand replies immediately.

5.6 Information Density

More information is available to both buyers and sellers. For customers: product comparisons, reviews, prices, delivery options, seller ratings. For businesses: customer behaviour, purchase history, location, preferences. This makes markets more transparent.

EXAMPLE

Before buying a phone, a customer can compare prices across many websites.

5.7 Personalization and Customization

Personalization means the system shows content based on user data (behaviour, name, interests, past purchases). **Customization** means the user changes the product or service according to their own

preference.

EXAMPLE

Netflix recommends movies based on your watch history (personalization). Choosing shoe colour before ordering is customization.

5.8 Social Technology

Users can create and share content: reviews, comments, photos, videos, product ratings, and social media posts. This enables user content generation and social networking.

EXAMPLE

A customer posts a review video of a product on TikTok. Others watch it and decide whether to buy.

6. Digital Markets

A **digital market** is an online market where buyers and sellers meet through the Internet. Digital markets reduce information asymmetry, search costs, transaction costs, and menu costs. They also allow dynamic pricing.

- Amazon
- Daraz
- Alibaba
- Upwork
- Fiverr
- Airbnb
- Uber
- eBay

7. Internet Reduces Information Asymmetry

> *Information asymmetry means one party in a transaction has more information than the other, which determines bargaining power.*

In traditional markets, a shopkeeper may know the actual quality and market price but the customer may not. E-commerce reduces this because customers can now check reviews, ratings, prices, product comparisons, seller reputation, and return policies.

EXAMPLE

Before buying a laptop, you can check prices on different websites, read reviews, and compare specifications. This reduces the seller's power to mislead you.

8. Digital Goods

Digital goods are goods that can be delivered over a digital network. Once a digital product has been produced, the cost of delivering another copy digitally is extremely low.

- Music tracks
- Videos / Movies
- Software

- Newspapers and Magazines
- Books (e-books)

> *Key insight: The first unit of a digital good has a high production cost, but the marginal cost of producing every additional unit is almost zero.*

EXAMPLE

Making a movie may cost millions. But after the movie is made, showing it to one more Netflix user costs almost nothing.

9. Disintermediation

Disintermediation means removing middlemen from the supply chain. E-commerce allows manufacturers to reach customers directly.

Traditional	Manufacturer → Distributor → Retailer → Customer
Partial	Manufacturer → Retailer → Customer
Full (E-Commerce)	Manufacturer → Customer

Benefits: Lower cost, faster delivery, more control over customer relationship, better profit margin.

EXAMPLE

A clothing brand sells directly through its website instead of using wholesalers and retailers.

10. Types of E-Commerce

B2C — Business-to-Consumer

Business sells directly to individual customers.

EXAMPLE

Amazon selling books, Daraz selling electronics, Netflix subscriptions, Foodpanda deliveries.

B2B — Business-to-Business

Business sells to another business. Helps companies locate suppliers, request bids, place orders, and track shipments electronically.

EXAMPLE

Alibaba suppliers selling to wholesalers, a software company selling CRM to another company.

C2C — Consumer-to-Consumer

Consumers sell to other consumers, usually through a marketplace platform.

EXAMPLE

OLX, Facebook Marketplace, eBay, a person selling a used phone online.

11. E-Commerce Business Models

Business Model	Description	Examples
E-Tailer	Online retail store that sells physical or digital products.	Amazon, Daraz, Alibaba retail stores
Transaction Broker	Platform that processes transactions for users and earns a fee per transaction.	Online ticket booking, stock trading apps, hotel booking websites
Market Creator	Creates a digital marketplace where buyers and sellers meet. Does not always own the product.	eBay, OLX, Airbnb, Upwork, Fiverr
Content Provider	Company that provides digital content.	Netflix, YouTube Premium, Spotify, online newspapers
Community Provider	Creates an online community. Value comes from user participation and interaction.	Facebook groups, Reddit, LinkedIn communities, online forums
Service Provider	Provides online services.	Google Drive, Canva, Dropbox, Zoom
Portal	Gives users a starting point to access many services including search, news, email, advertising, and links.	Google, Yahoo, MSN

12. E-Commerce Revenue Models

Revenue Model	How it Works	Examples
Advertising	Earns money by showing ads.	YouTube, Facebook, Google, news websites
Sales	Earns money by selling goods or services directly.	Amazon, Daraz, Apple
Subscription	User pays a recurring fee for access.	Netflix, Spotify Premium, Coursera, online newspapers
Free / Freemium	Basic version is free; advanced features are paid.	Canva, Zoom, Dropbox, Grammarly, ChatGPT plans
Transaction Fee	Platform earns money per completed transaction.	Upwork, Airbnb, payment gateways, booking platforms
Affiliate	Earns commission by sending customers to another business.	A blogger adds Amazon affiliate links; earns commission on purchases

13. Behavioral Targeting

Behavioral targeting means showing ads based on user behaviour. It increases the effectiveness of banner ads, rich media ads, and video ads.

- Search history
- Pages visited
- Products viewed
- Items added to cart
- Videos watched
- Past purchases

EXAMPLE

You search for shoes online. Later, shoe ads appear on Facebook, YouTube, and other websites.

14. Social Network Marketing

Uses user relationships, likes, shares, comments, and communities to promote products.

Social Graph

A social graph shows relationships between people online. Facebook knows who your friends are, what pages you like, and what content you engage with.

Wisdom of Crowds

Many people together can provide useful collective opinions. Social networking and the wisdom of crowds help companies improve marketing and product offerings.

EXAMPLE

Thousands of customer reviews on Amazon help new buyers decide.

15. Case Studies: Netflix and Daraz

Netflix

Concept	Netflix Example
E-commerce	Sells entertainment through the Internet
Digital goods	Movies and shows
Subscription model	Monthly payment
Personalization	Recommends shows based on watch history
Global reach	Available in many countries
Ubiquity	Can watch anytime, anywhere
Low marginal cost	One more user watching costs very little
Social impact	People discuss shows online

Daraz

Concept	Daraz Example
E-commerce type	B2C and C2C marketplace
Business model	Market creator and e-tailer
Revenue model	Sales commission, ads, transaction fees
Ubiquity	Customers can buy from mobile anytime
Information density	Reviews, ratings, prices, product details
Interactivity	Chat with seller
Personalization	Product recommendations
Digital market	Online marketplace connecting buyers and sellers

16. Most Important Exam Topics

- Definition of e-commerce
- Difference between traditional commerce and e-commerce
- Social, mobile, and local e-commerce
- Online marketing methods
- Eight unique features: Ubiquity, Global Reach, Universal Standards, Richness, Interactivity, Information Density, Personalization/Customization, Social Technology
- Digital markets and how they differ from traditional markets
- Internet reduces information asymmetry
- Digital goods and marginal cost
- Intellectual property in digital goods
- Disintermediation
- Types of e-commerce: B2C, B2B, C2C
- E-commerce business models (7 types)
- E-commerce revenue models (6 types)
- Behavioral targeting
- Social network marketing: Social Graph, Wisdom of Crowds
- E-commerce presence: Website, Email, Social media, Offline support

Chapter 10 explains how the Internet creates digital markets where businesses sell physical and digital goods globally, cheaply, interactively, and personally through websites, mobile apps, and social platforms.